

illustrated in Figure 3 (2), this system condenses the original text into a compact yet informative paragraph, thereby enhancing FINMEM’s processing efficiency. It efficiently extracts and summarizes pertinent data and sentiments for stock investment decisions, demonstrated here using Tesla Inc. as an example. Subsequently, FINMEM directs these insights to an appropriate layer within its long-term memory architecture, selecting the layer based on the time sensitivity of the information.

Observation: Triggered the same inquiry, FINMEM initiates an observation operation to gather market facts. The information available to FINMEM varies between the training and testing phases.

During the training phase, FINMEM has access to comprehensive stock price data within the specified period. Upon receiving trading inquiries that specify a stock ticker and date, FINMEM focuses on the daily adjusted closing price differences, comparing the following day’s price with the current day’s. These price differences are utilized as market ground labels. Specifically, a decrease in price suggests a “Sell” action, while an increase or no change in price indicates a “Buy” action.

During the testing phase, at a specific time point, FINMEM loses the ability to access future price data. Its focus shifts to the analysis of historical stock price movements, depending on a retrospective evaluation of the cumulative return from the last M trading days to infer future market trends. This phase, characterized by the absence of foreseen market grounds, serves as a critical assessment of FINMEM’s development. It tests whether the system has adequately established logical connections between stock price trends and various financial information sources, such as news, reports, and indicators. This stage is key in evaluating FINMEM’s capability of independently evolving its trading strategies for subsequent tasks, leveraging its analysis and interpretation of historical data patterns.

Reflection: Two types of reflections exist, immediate and extended reflection. (a) Immediate reflection is activated upon receiving a daily trading inquiry for a specific ticker. Using LLM and specific prompts exemplified in Figure 3 (2), the agent merges market indications and top- K -ranked events from each long-term memory layer. Market indications are derived from the outcomes of the observation operation and differ between the training and testing phases. During testing, this process yields three types of outputs: the trading direction (“Buy”, “Sell”, or “Hold”), the underlying rationale for this decision, and the most influential memory events, along with their IDs from each layer that informed the decision. In the training phase, specifying the trading direction is unnecessary, as FINMEM is already informed of future stock movement directions. The top- K -ranked memory events encapsulate key insights and sentiments derived from critical investment-related incoming messages, all distilled by FINMEM’s advanced summarization capabilities.

(b) Extended reflection reevaluates immediate reflection outcomes for a ticker over a specified M -day trace period. It encompasses data like stock price trends, trading returns, and action rationales from multiple immediate reflections. While immediate reflection enables direct trading execution and records current feedback, extended reflection summarizes market trends and reassesses recent Cumulative Return on investment. Extended reflection is eventually transmitted and stored in the deep processing layer to emphasize its criticality (detailed introduced in Section 3.2.2) of long-term memory. K and M are hyperparameters to adjust FINMEM’s working memory capacity and information retrieval ability. FINMEM gains the flexibility of integrating comprehensive information into well-informed decisions by fine-tuning them.

3.2.2 Layered long-term memory

FINMEM’s long-term memory organizes hierarchical financial data insights in a stratified structure, as illustrated in the lower section of Figure 2. Drawing inspiration from the varying decay speeds in the human cognitive system’s information processing layers [9], FINMEM employs a layered structure to accommodate the diverse time sensitivities inherent to different types of financial data. This structure categorizes summarized insights by their timeliness and decay rates. Insights are derived by the working memory’s summarization operation. Those directed to deeper layers receive smaller decay rates, indicating longer retention, while those in shallower layers are assigned larger decay rates for shorter retention.

$$\gamma_l^E = S_{\text{Recency}_l}^E + S_{\text{Relevancy}_l}^E + S_{\text{Importance}_l}^E, \quad (1)$$

where each memory event is only associated with one score and can only belong to a single layer.

Upon receiving an investment inquiry, FINMEM retrieves the top- K pivotal memory events from each layer and channels them to the immediate reflection component of the working memory. These events are chosen according to the descending order of their information retrieval score, denoted as γ_l^E , where l belongs to the set shallow, intermediate, deep, as specified in Equation 5. E denotes a given memory event. This score, adapted from Park et al. [35] but with modified recency and importance computations, especially tailoring to handle data with various timelines. It

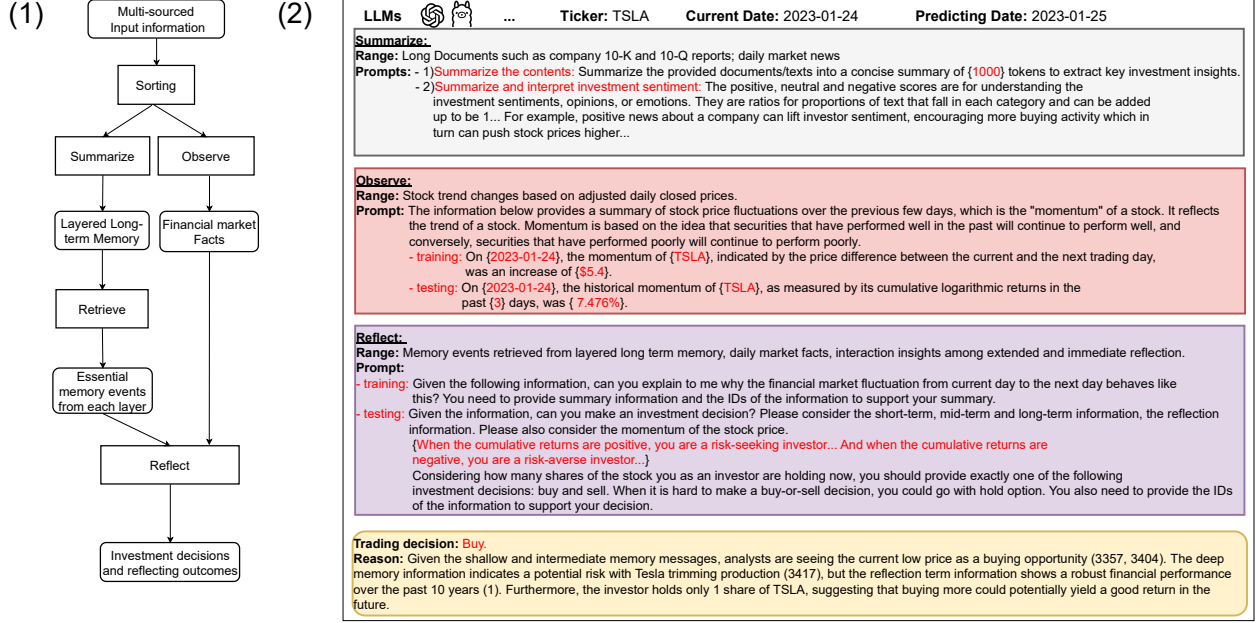


Figure 3: (1) The decision-making module workflow of the FINMEM trading agent retrieves critical memory events to inform specific decisions. (2) LLM prompt template used by FINMEM to interact with incoming financial information.

encapsulates three metrics: recency, relevancy, and importance. Individual metric scores exceeding 1.0 are scaled to the [0,1] range before being summed. The modification is to achieve the layered processing function and represent the various periodicity of the financial environment.

$$S_{\text{Recency}_l}^E = e^{-\frac{\delta^E}{Q_l}}, \quad \delta^E = t_P - t_E, \quad (2)$$

where δ^E refers to the time difference between the memory event occurrence and the trading inquiry arrival. $Q_{\text{shallow}} = 14$, $Q_{\text{intermediate}} = 90$, and $Q_{\text{deep}} = 365$ correspond to day counts of two weeks, a quarter, and a year for shallow, intermediate, and deep processing layers, respectively.

Upon a trade inquiry P arrival in processing layer l via LLM prompt, the agent computes the recency score $S_{\text{Recency}_l}^E$ per Equation.2. $S_{\text{Recency}_l}^E$ inversely correlates with the time gap between the inquiry and the event's memory timestamp, mirroring Ebbinghaus's forgetting curve [33]. The stability term Q_l in Equation.2 partially controls memory decay rates across layers, indicating longer memory persistence in the long-term layer with a higher stability value. In the context of trading, company annual reports, such as Form 10-Ks, are considered to have more extended timeliness compared to daily financial news. Therefore, they are assigned a higher stability value and are categorized within the deeper processing layer. This classification reflects their extended relevance and impact in financial decision-making scenarios.

$$S_{\text{Relevancy}_l}^E = \frac{\mathbf{m}_E \cdot \mathbf{m}_P}{\|\mathbf{m}_E\|_2 \times \|\mathbf{m}_P\|_2} \quad (3)$$

The relevancy score, denoted as $S_{\text{relevancy}_l}^E$, quantifies the cosine similarity between the embedding vectors. These vectors are derived from the textual content of the memory event, $\mathbf{m}_E$, and the LLM prompt query, $\mathbf{m}_P$, using OpenAI's "text-embedding-ada-002" model, as depicted in Equation 3. The LLM prompt query incorporates inputs related to trading inquiries and the trading agent's character setting.

The importance score $S_{\text{Importance}_l}^E$ is computed using the value v_l^E from a uniform piecewise scoring function (Formula 4), multiplied by a degrading ratio θ_l (Formula 5) as per Equation 6. The likelihood of higher v_l^E values increases from shallow to deep layers. θ_l measures the diminishing importance of an event over time, which has a close form design of [35]. But our approach tailors θ_l to the stratified structure of long-term memory. It adopts unique exponential functions for each layer. The base α_l for each layer is a hyperparameter, set to follow the sequence:

$\alpha_{\text{shallow}} < \alpha_{\text{intermediate}} < \alpha_{\text{deep}}$. These values correlate with the rate at which their importance degrades after a certain period, providing another angle to measure importance variances across different memory types. Through experimentation, we set $\alpha_{\text{shallow}} = 0.9$, $\alpha_{\text{intermediate}} = 0.967$ and $\alpha_{\text{deep}} = 0.988$. This ensures that θ_l decreases to a threshold score of 5 after intervals of 30, 90, and 365 days for shallow, intermediate, and deep layers, respectively. The three-piece-wise functions for $S_{\text{Importance}_l}^E$ and $S_{\text{Recency}_l}^E$ enable FINMEM to have layered processing in the long-term memory component. Memory events are purged when $S_{\text{Recency}_l}^E$ is below 0.05 or $S_{\text{Importance}_l}^E$ is under 5 (pre-scaling).

$$v_l^E = \begin{cases} 40 & \text{with probability } p_1 \\ 60 & \text{with probability } p_2 \\ 80 & \text{with probability } p_3 \end{cases} \quad (4)$$

$$\theta_l = (\alpha_l)^{\delta^E}, \quad l = \text{shallow, intermediate, deep}, \quad (5)$$

where $p_1 + p_2 + p_3 = 1$, but their values vary by shallow, intermediate, and deep processing. when shallow processing $p_1, p_2, p_3 = \{0.8, 0.15, 0.05\}$, intermediate processing, $p_1, p_2, p_3 = \{0.05, 0.8, 0.15\}$ and deep processing, $p_1, p_2, p_3 = \{0.05, 0.15, 0.8\}$.

$$S_{\text{Importance}_l}^E = v_l^E * \theta_l, \quad (6)$$

Furthermore, an access counter function oversees the transfer of memory events among layers, ensuring that significant events influencing trading decisions ascend from shallower to deeper layers for extended retention and recurrent access by FINMEM. Conversely, less pertinent events gradually diminish. This process is facilitated by the LLM validation tool Guardrails AI [17], which monitors critical memory IDs across different layers. An event identified as pivotal for investment success receives an additional 5 points in its importance score $S_{\text{Importance}_l}^E$. Upon meeting the criteria for upgrading to a deeper layer, an event’s recency score $S_{\text{Recency}_l}^E$ is reset to 1.0, emphasizing its importance and preventing rapid decay. By implementing this access counter, FINMEM effectively identifies and prioritizes key events, taking into account their nature and frequency of retrieval.

3.3 Decision-making Module

The decision-making module of FINMEM efficiently integrates operational outcomes from the profiling and memory modules to support well-informed investment decisions, as depicted in Figure 3 (1). In its daily trading decisions, FINMEM is asked to select from three distinct actions for a single share of a specific stock by Guardrails AI text validation function: “Buy”, “Sell”, or “Hold”. Additionally, the inputs and results required by FINMEM’s decision-making module vary between its training and testing phases, with each phase’s specifics detailed as follows:

During the training phase, FINMEM accesses a wide array of multi-source information relevant to the entire time period. When FINMEM is prompted with trading inquiries containing stock ticker and date, as well as trader character-related texts, it concurrently initiates observation and summarization operations in its working memory. FINMEM observes the market ground labels mentioned in the description about the observation operation in Section 3.2.1, which involve daily adjusted price differences between consecutive days, indicative of “Buy” or “Sell” actions. Utilizing these price change signals, FINMEM identifies and prioritizes the top- K memories, ranking them based on retrieval scores from each long-term memory layer. This procedure enables FINMEM to produce comprehensive reflections that provide a well-founded rationale and in-depth inference of the correlation between market ground labels and the memories retrieved. Through repeated trading operations, reflections, and memory events with significant impact, transition to a deeper memory processing layer, getting preserved for guiding future investment decisions during the testing phase.

In the testing phase, where FINMEM cannot access future price data, it relies on the Cumulative Return over the previous M trading days to anticipate future market trends. To compensate for the absence of future market price information, FINMEM utilizes enhanced reflections derived from immediate reflections spanning an M -trading-day period as supplementary references. When faced with a specific trading inquiry, FINMEM integrates insights from various sources, including historical Cumulative Return, outcomes from extended reflection, and the Top- K retrieved memories. This comprehensive approach enables FINMEM to execute well-informed trading decisions.

It should be noted that FINMEM generates executable actions exclusively in the immediate reflection operation of the testing phase. Since the trading direction is guided by the actual price trend, the training phase of FINMEM does not make investment decisions. Instead, this phase is dedicated to accumulating trading experience through comparing